

Overview

Tony Feng

Representation theory

$$\underline{G} = \text{split red. gp} / \mathbb{A}_p$$

Let $k = \text{coeff. ring}$, $p \in k^\times$, e.g. $\mathbb{A}_\ell, \mathbb{Z}_\ell \subset \mathbb{F}_\ell$

Interested in reps of $\underline{G}(\mathbb{A}_p) = G$ on k -modules which are smooth.

i.e. $\forall x \in V, \exists U \subset \underline{G}(\mathbb{A}_p)$ cpt open s.t. U fixes x

More conceptually, it means the action map $\underline{G}(\mathbb{A}_p) \times V \rightarrow V$ is cts (V w/ discrete top)

Hecke algebras

Let U open cpt $\subset \underline{G}(\mathbb{A}_p)$

$$\mathcal{H}(G, U) = \text{Func}_{\substack{\uparrow \\ \text{cpt supp}}} (U \backslash \underline{G}(\mathbb{A}_p) / U(k)) \quad \text{associative alg under convolution}$$

Fact If V is a $\underline{G}(\mathbb{A}_p)$ -rep, then V^U has a canonical $\mathcal{H}(G, U)$ -action.

Explicitly, $f \in \mathcal{H}(G, U)$, $x \in V^U$,

$$f * x = \int_G f(g) g \cdot x \, dg \in V^U$$

More abstractly, let $k[G/U] = \text{Func}^\infty(G/U; k)$, then $\mathcal{H}(G, U) = \text{End}_G(k[G/U])$

acts tautologically on $\text{Hom}_k(k[G/U], V) = \text{Hom}_U(k, V) = V^U$.

Def'n $Z(G, U) = \text{centre of } H(G, U)$

Ex $U = G(\mathbb{Z}_p) (= K)$. Then $H(G, U)$ is cplly supp func. on

$$K \backslash G(\mathbb{Q}_p) / K \longleftrightarrow X_*(T)^+$$

Satake transf.

In this case, $H(G, K)$ is actually commutative. $\xrightarrow{\sim} k[X_*(T)]^W$ * chosen $\sqrt{q}$

Ex $U = I$ Iwahori $\rightarrow G(\mathbb{Z}_p)$

$$\begin{array}{ccc} & \downarrow & \downarrow \\ B(\mathbb{F}_p) & \rightarrow & G(\mathbb{F}_p) \end{array}$$

$$H(G, I) \stackrel{k\text{-mod}}{=} \text{Func}(I \backslash G(\mathbb{Q}_p) / I) \longleftrightarrow \tilde{W} \text{ extended affine Weyl gp}$$

$\uparrow$

$$X_*(T) \rtimes W$$

"lattice part"

Turns out

$$H(G, I) = \langle H_x : x \in \tilde{W} \mid H_x H_y = H_{xy}, \ell(xy) = \ell(x) + \ell(y) \rangle$$

simple

$$H_s^2 = H_{Id} + (q^{-1} - q) H_s \quad \text{if } s \text{ is a reflection} \rangle$$

$\cup$

$$k[X_*(T)]^{\text{comm.}} \quad \lambda \in X_*(T) \quad \text{write } \lambda = \mu - \mu', \quad \mu, \mu' \in X_*(T)^+$$

$$\begin{array}{ccc} \uparrow & \cup & \\ A_\lambda & & A_\lambda = H_\mu (H_{\mu'})^{-1} \\ & k[X_*(T)]^W & \text{central} \end{array}$$

(Toy model: $|w|x|w|$ mat. alg / centre)

$\mathcal{H}(G, I)$ is a q -deformation of $k[\tilde{W}] \supset k[X_*(T)] \supset k[X_*(T)]^W$

heveral features

- $\mathcal{H}(G, U)$ is complicated
- I_+ has a large central subalg. (finite over its centre)
- Central elts hard to write down

Intuition $\mathcal{Z}(G, U)$ is close to ring of functions on the 'corresponding' part of moduli of Galois rep's.

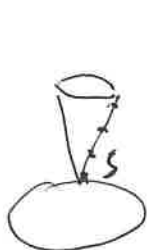
$$\underline{\text{Ex}} \quad U = K = G(\mathbb{Z}_p) \longleftrightarrow \left\{ \begin{array}{l} \text{unramified part} \\ W_{\text{clp}} \twoheadrightarrow \mathbb{Z} \rightarrow \hat{G}^{(\text{sq})}(K) \end{array} \right\}$$

$$\hat{G} / \hat{G}^{\text{conj.}}$$

$$\text{Functions } \mathcal{O}(\hat{G})^{\hat{G}} \xrightarrow{\text{Chevalley}} k[X^*(T)]^W = k[X_*(T)]^W$$

$$= \mathcal{H}(G, K)$$

$$\underline{\text{Ex}} \quad U = I \longleftrightarrow \text{tamely ramified (unipotent monodromy)}$$



$$\left\{ W_{\text{clp}} \twoheadrightarrow I \rtimes \mathbb{Z} \rightarrow \hat{G}(K) \right\}$$

$$\text{Functions } \mathcal{O}(\hat{G})^{\hat{G}} = k[X_*(T)]^W$$

$$\left\{ (x \mapsto \hat{g}, s \in \hat{G}) : sxs^{-1} = qx \right\} / \hat{G} \xrightarrow{\text{Bernstein}} \mathcal{Z}(\mathcal{H}(G, I))$$

Main Thm of Paper $k = \text{noetherian } \mathbb{Z}_\ell\text{-alg. } (\mathbb{Z}[\frac{1}{p}])$

- $\mathcal{H}(G, U)$ is finite over $\mathcal{Z}(G, U)$

- $\mathcal{Z}(G, U)$ is finitely gen. $/k$.

Remark If $k = \mathbb{C}$, then this is old result of Bernstein.

Rep theory implications $P = MN$ parabolic $\subset G$

$\Rightarrow$ parabolic induction $\text{Ind}_P^G: \text{Rep}_k M(\mathcal{O}_P) \rightarrow \text{Rep}_k G(\mathcal{O}_P)$

$$\text{Ind}_P^G(V) = \{ \text{smooth } f: G(\mathcal{O}_P) \rightarrow V : f(hg) = h \cdot f(g), \forall h \in P(\mathcal{O}_P) \}$$

normalized version $i_P^G(V) = \text{Ind}_{P(\mathcal{O}_P)}^{G(\mathcal{O}_P)} (\delta_P^{1/2} V)$

$$\uparrow$$

$$\delta_P = \det(M \sim N)$$

Def. V is admissible if $\dim V^U < \infty$ for all open cpt $U \subset G$.

Compactness of $P(\mathcal{O}_P) \backslash G(\mathcal{O}_P) \xRightarrow{\text{exer}} \left[\begin{array}{l} i_P^G \text{ preserves admissibility and fin. gen.} \\ \Gamma_P^G \text{ preserves admissibility and fin. gen.} \end{array} \right]$

$\leftarrow \text{Main Thm} \rightarrow$

eg. Let $U \subset^{\text{cpt open}} G$

$P(\mathcal{O}_P) \backslash G(\mathcal{O}_P) / U$ is finite, say size 2

$$\Rightarrow G(\mathcal{O}_P) = P(\mathcal{O}_P) g_1 U \cup P(\mathcal{O}_P) g_2 U$$

$$(\text{Ind}_P^G V)^U \xrightarrow{\text{eval at } g_1, g_2} V \times V$$

$\xrightarrow{U_M \times U_N} V \times V$

some open cpt $U'_M \subset M$

$$U' = g_1^{-1} U g_1 \cap g_2^{-1} U g_2, \quad U'_M = \frac{\text{Im}}{U' \cap \mathfrak{p}} \rightarrow M$$

$$f(g_1, U'_M) = f(g_1)$$

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$$f(g_1, g_1^{-1} U g_1, \dots)$$

of the form smoothly open g_1

Bernstein Centre

$$\mathcal{Z}(G) = \varprojlim_{\substack{U \\ \text{cpt open } U \subseteq G \\ \text{pro-p}}} \mathcal{Z}(G, U)$$

$$U' \in U$$

$$\mathcal{Z}(G, U') \xrightarrow{*1_U} \mathcal{Z}(G, U)$$

is the Bernstein centre of $\text{Rep } G(\mathbb{A}_f)$

(= alg. of compatible endomorphisms of all objects)

$$V \text{ smooth} \Leftrightarrow V = \varinjlim_U V^U$$

$$\mathcal{Z}(G) = \varprojlim_U \mathcal{Z}(G, U)$$

Def'n Say $V \in \text{Rep}_k G$ is $\mathcal{Z}$ -finite if

- $\text{Im}(\mathcal{Z}(G) \rightarrow \text{End}_G(V))$ is fin. gen. / k

- V is admissible over $\mathcal{Z}(G)$: V^U is finite over $\mathcal{Z}(G)$, $\forall$ open cpt $U \subseteq G$.

Rank Closed under passage to subquotients.

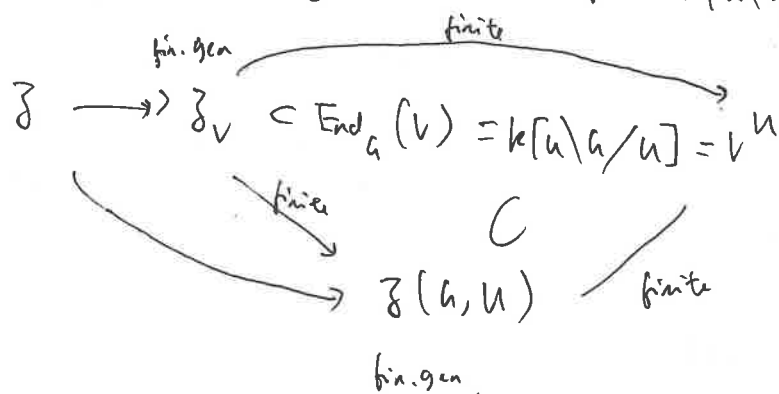
Lemma Let k noeth, $p \in k^\times$. TFAE

(1) $\forall$ open cpt U , $H(G, U)$ finite over $\mathbb{Z}(G, U)$ and $\mathbb{Z}(G, U)$ t -gen. / k

(2) Any fin. gen. G -reph is $\mathbb{Z}$ -finite.

Proof (2) $\Rightarrow$ (1) Let $V = k[G/U]$ fin. gen.

hyp (2) $\Rightarrow V$ is $\mathbb{Z}$ -finite $\Rightarrow V^U = k[U \backslash G/U]$ finite / $\mathbb{Z}$



Remark We will apply this to "large" V

$$\chi^{\text{univ}} : T(\mathbb{Q}_p) / T(\mathbb{Z}_p) = X_*(\Gamma) \rightarrow k[X_*(\Gamma)]$$

$$V = \text{Ind}_B^G (\chi^{\text{univ}})$$

$\text{End}_G(V)$ needs to have a large centre.

• Bernstein proved this in char $k=0$.

Toy model Let $A = \text{assoc. alg.}$ Then

$$\mathbb{Z}(A) = \mathbb{Z}(A\text{-mod}) = \{ \text{compatible endom. of all } A\text{-modules} \}$$

$$A = k[\Gamma] \quad \text{finite group } \Gamma.$$

$$k = \mathbb{C} \Rightarrow \mathbb{C}[\Gamma] = \prod_{V \text{ irrep}} \text{End}(V). \quad \mathbb{Z}(\mathbb{C}[\Gamma]) = \prod_{V \text{ irrep}} \mathbb{Z}$$

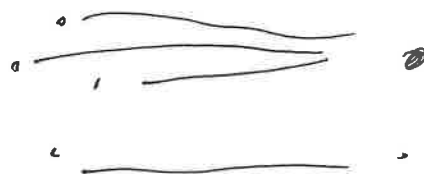
If $\text{char } k = \ell > 0$, then it looks very different.

eg. $k = \mathbb{F}_\ell$, $\mathbb{F} = \mathbb{Z}/\ell$.

$\text{char } k = 0$

$\text{char } k = \ell$

$$k[\tau] = \mathbb{F}_\ell[t]/t^{\ell-1} = \mathbb{F}_\ell[s]/s^\ell$$



Bernstein decomposition $k = \mathbb{C}$

$$\text{Rep } G(\mathbb{Q}_p) = \prod_{[M, \sigma]} \text{Rep } G[M, \sigma]$$

$M = \text{Levi}$, $\sigma = \text{supercuspidal rep'n}$, $\sim = \text{unramified twist, conjugation}$
i.e. $\text{char of } M/M_0$

where $M_0 = \text{union of cpt open subgrps}$

eg. $M = \text{GL}_n(\mathbb{Q}_p)$, then $M_0 = \text{subgp } \forall \det \in \mathbb{Z}_p^\times$

$\text{Rep } G[M, \sigma]$ is pro-generated by $i_p^G(\sigma \otimes \mathbb{C}[M/M_0])$

$$\mathcal{O} \left(\mathbb{C}_M \otimes (M/M_0) / \text{stab}(\sigma) \right)^{\text{Weyl}}$$

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centre of the block

$$\text{End}_G(i_p^G(\sigma \otimes \mathbb{C}[M/M_0]))$$

$\uparrow$

$$\text{End}(\sigma \otimes \mathbb{C}[M/M_0])$$

If $\text{char } k \neq 0$, lose Bernstein theory.
instead use LLC (Fargues-Scholze)

Intuition. centres of Hecke alg look like ring of fractions on moduli of

Hecke reps $\{ \rho: W_{\mathbb{A}_F} \xrightarrow{\text{cts}} \hat{G} \} / \hat{G} = M_{\hat{G}}$

excursion algebra $O(\mathcal{M}_{\hat{G}}) \xrightarrow{FS_{\hat{G}}} \mathcal{Z}(\hat{G}) \hookrightarrow \pi$
 $\text{Exc}(W_F, \hat{G}) \longrightarrow \pi$

- good finiteness properties from geometry of $M_{\hat{G}}$

- characters $\text{Exc}(W_F, \hat{G}) \rightarrow k$

$\Downarrow$

semisimple $\rho: W_{\mathbb{A}_F} \xrightarrow{\text{cts}} \hat{G}(k)$

- compatibility of FS w parabolic induction, etc.

Repeat Bernstein argument w $\mathcal{Z}$ replaced by Exc . and use finiteness:

$$\text{Exc}(W_F, \hat{G}) \xrightarrow{\text{red}} \text{Exc}(W_F, \hat{G})_{\text{red}} \text{ finite}$$

$$\{ W_{\mathbb{A}_F} \rightarrow \hat{G} \} / \hat{G} \text{ finite} \leftarrow \{ W_{\mathbb{A}_F} \rightarrow \hat{G} \} / \hat{M}$$

Ex. $\hat{M} = \hat{T} \quad \begin{pmatrix} x_1 & \\ & x_2 \end{pmatrix} \quad \begin{pmatrix} x_2 & \\ & x_1 \end{pmatrix}$

$\searrow \swarrow$